

PREVIOUS YEARS'

CBSE BOARD QUESTIONS

1.1 Introduction

[NCERT Page 1]

MCQ

- $(3 \times 11 \times 13 + 3)$ is
(a) a prime number
(b) divisible by 13
(c) a composite number
(d) an odd number
(2026)
- The natural number 1 is
(a) a prime number
(b) a composite number
(c) prime as well as composite
(d) neither prime nor composite
(2026) R
- The natural number 2 is :
(a) a prime number
(b) a composite number
(c) prime as well as composite
(d) neither prime nor composite
(2026) R
- If $(-1)^n + (-1)^8 = 0$, then n is :
(a) any positive integer
(b) any negative integer
(c) any odd number
(d) any even number
(2025) An
- The total number of factors of a prime number is
(a) 1 (b) 0 (c) 2 (d) 3 (2020) R



1.2 The Fundamental Theorem of Arithmetic

[Page 2-6]

MCQ

- Assertion (A) :** $H.C.F.(36m^2, 18m) = 18m$, where m is a prime number.
Reason (R) : H.C.F. of two numbers is always less than or equal to the smaller number.
(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.
(2026)
- There are two sections A and B of Grade X. There are 28 students in Section A and 30 students in Section B. What is the minimum number of books you will acquire for the class library so that they can be

distributed equally among students of Section A or Section B?

- (a) 144 (b) 2 (c) 420 (d) 272 (2026) Ap
- For any natural number n , 5^n ends with the digit
(a) 0 (b) 5 (c) 3 (d) 2 (2026) U
- The LCM of 960 and 240 is :
(a) 960 (b) 240 (c) 60 (d) 15 (2026) Ap
- The HCF of 960 and 432 is :
(a) 48 (b) 54 (c) 72 (d) 36 (2026) Ap
- For any natural number n , 6^n ends with the digit:
(a) 0 (b) 6
(c) 3 (d) 2 (2026) U
- The total number of factors of the square of a prime number is :
(a) 1 (b) 2 (c) 3 (d) 4 (2025 C) U
- Which of the following cannot be the unit digit of 8^n , where n is a natural number?
(a) 4 (b) 2
(c) 0 (d) 6 (2025) An
- If x is the LCM of 4, 6, 8 and y is the LCM of 3, 5, 7 and p is the LCM of x and y , then which of the following is true?
(a) $p = 35x$ (b) $p = 4y$
(c) $p = 8x$ (d) $p = 16y$ (2025) Ap
- If $HCF(98, 28) = m$ and $LCM(98, 28) = n$, then the value of $n - 7m$ is :
(a) 0 (b) 28
(c) 98 (d) 198 (2025) Ev
- The greatest number which divides 70 and 125, leaving remainders 5 and 8 respectively, is :
(a) 13 (b) 65
(c) 875 (d) 1750 (2025) U
- Assertion (A) :** For any two prime numbers p and q , their HCF is 1 and LCM is $p + q$.
Reason (R) : For any two natural numbers, $HCF \times LCM = \text{product of numbers}$.
(a) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.
(2025) Ap

18. Two positive integers m and n are expressed as $m = p^5 q^2$ and $n = p^3 q^4$, where p and q are prime numbers. The LCM of m and n is:
 (a) $p^8 q^6$ (b) $p^3 q^2$
 (c) $p^5 q^4$ (d) $p^5 q^2 + p^3 q^4$ (2024C) 
19. If two positive integers p and q can be expressed as $p = 18 a^2 b^4$ and $q = 20 a^3 b^2$, where a and b are prime numbers, then LCM (p, q) is:
 (a) $2a^2 b^2$ (b) $180a^2 b^2$
 (c) $12 a^2 b^2$ (d) $180 a^3 b^4$ (2024) 
20. If $a = 2^2 \times 3^x$, $b = 2^2 \times 3 \times 5$, $c = 2^2 \times 3 \times 7$ and LCM (a, b, c) = 3780, then x is equal to
 (a) 1 (b) 2
 (c) 3 (d) 0 (2024) 
21. Given HCF (2520, 6600) = 40, LCM (2520, 6600) = $252 \times k$, then the value of k is:
 (a) 1650 (b) 1600
 (c) 165 (d) 1625 (2024) 
22. The LCM of the smallest prime number and the smallest odd composite number is:
 (a) 10 (b) 6
 (c) 9 (d) 18 (2024) 
23. The greatest number which divides 281 and 1249, leaving remainder 5 and 7 respectively, is:
 (a) 23 (b) 276
 (c) 138 (d) 69 (2024) 
24. The LCM of three numbers 28, 44, 132 is:
 (a) 258 (b) 231
 (c) 462 (d) 924 (2024) 
25. **Assertion (A)** : The number 5^n cannot end with the digit 0, where n is a natural number.
Reason (R) : Prime factorisation of 5 has only two factors, 1 and 5.
 (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of the Assertion (A).
 (c) Assertion (A) is true but Reason (R) is false.
 (d) Assertion (A) is false but Reason (R) is true. (2023) 
26. If ' p ' and ' q ' are natural numbers and ' p ' is the multiple of ' q ', then what is the HCF of ' p ' and ' q '?
 (a) pq (b) p
 (c) q (d) $p + q$ (2023) 
27. The ratio of HCF to LCM of the least composite number and the least prime number is
 (a) 1 : 2 (b) 2 : 1
 (c) 1 : 1 (d) 1 : 3 (2023)

28. If HCF (39, 91) = 13, then LCM (39, 91) is
 (a) 91 (b) 273 (c) 39 (d) 3549
 (Term I, 2021-22) 
29. Two positive numbers have their HCF as 12 and their product as 6336. The number of pairs possible for the numbers, is
 (a) 2 (b) 3 (c) 4 (d) 1
 (Term I, 2021-22) 
30. If ' n ' is any natural number, then $(12)^n$ cannot end with the digit
 (a) 2 (b) 4 (c) 8 (d) 0
 (Term I, 2021-22)
31. The number 385 can be expressed as the product of prime factors as
 (a) $5 \times 11 \times 13$ (b) $5 \times 7 \times 11$
 (c) $5 \times 7 \times 13$ (d) $5 \times 11 \times 17$
 (Term I, 2021-22)
32. The HCF and the LCM of 12, 21 and 15 respectively, are
 (a) 3, 140 (b) 12, 420
 (c) 3, 420 (d) 420, 3 (2020) 

VSA (1 mark)

33. The LCM of two numbers is 182 and their HCF is 13. If one of the numbers is 26, find the other. (2020)
34. The LCM of two numbers is 9 times their HCF. The sum of LCM and HCF is 500. Find the HCF of the two numbers. (2019C) 
35. If HCF (336, 54) = 6, find LCM (336, 54). (2019) 
36. The HCF of two numbers a and b is 5 and their LCM is 200. Find the product ab . (AI 2019)
37. What is the HCF of smallest prime number and the smallest composite number? (2018)
38. Show that any number of the form 6^n , where $n \in \mathbb{N}$ can never end with digit 0. (Board Term I, 2017) 
39. The HCF of two numbers is 27 and their LCM is 162, if one of the number is 54, find the other number. (Board Term I, 2017) 

SA I (2 marks)

40. Find the length of the plank that can be used to measure the lengths 4 m 20 cm and 5 m 4 cm exactly, in the least time. (2026) 
41. If LCM of 51 and 85 can be expressed in the form $6z - 9$, then find the value of z . (2025C) 
42. Find the smallest number which is divisible by both 644 and 462. (2025) 

43. Two numbers are in the ratio 4 : 5 and their HCF is 11. Find the LCM of these numbers. (2025)  **Ap**
44. Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number. (2024)  **An**
45. Check whether 6^n can end with the digit 0 for any natural number n . (2023C)  **An**
46. Find the least number which when divided by 12, 16 and 24 leaves remainder 7 in each case. (2023)
47. Two numbers are in the ratio 2 : 3 and their LCM is 180. What is the HCF of these numbers? (2023)
48. Explain why $2 \times 3 \times 5 + 5$ and $5 \times 7 \times 11 + 7 \times 5$ are composite numbers. (2021C)  **U**
49. If HCF of 65 and 117 is expressible in the form $65n - 117$, then find the value of n . (2019)
50. Find the HCF of 612 and 1314 using prime factorisation. (AI 2019)  **Ap**
51. Express 5050 as product of its prime factors. Is it unique? (Board Term I, 2017)
52. Show that the numbers 231 and 396 are not co-prime. (Board Term I, 2017)

SA II (3 marks)

53. The dimensions of a window are $156 \text{ cm} \times 216 \text{ cm}$. Arjun wants to put grill on the window creating complete squares of maximum size. Determine the side length of the square and hence find the number of squares formed. (2026)  **Ev**
54. Ranjita, Neha and Salma start weaving sweaters at the same time for the children of an orphan home. They need 15, 18 and 20 days, respectively, to complete a sweater. After how many days will all of them start making a new sweater again? By that time how many sweaters will have been completed by them? (2025C)  **Ev**
55. Three sets of Physics, Chemistry and Mathematics books have to be stacked in such a way that all the books are stored subject-wise and the height of each stack is the same. The number of Physics books is 144, the number of Chemistry books is 180 and the number of Mathematics books is 192. Assuming that the books are of same thickness, determine the number of stacks of Physics, Chemistry and Mathematics books. (2025)  **Cr**
56. Let p, q and r be three distinct prime numbers. Check whether $p \cdot q \cdot r + q$ is a composite number or not. Further, give an example for 3 distinct primes p, q, r such that
- $p \cdot q \cdot r + 1$ is a composite number.
 - $p \cdot q \cdot r + 1$ is a prime number.
- (2025)  **Cr**
57. A school has invited 42 Mathematics teachers, 56 Physics teachers and 70 Chemistry teachers to attend

a Science workshop. Find the minimum number of tables required, if the same number of teachers are to sit at a table and each table is occupied by teachers of the same subject. (2024C)  **Cr**

58. In a teacher's workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject. (2024)  **Cr**
59. Find by prime factorisation the LCM of the numbers 18180 and 7575. Also, find the HCF of the two numbers. (2023)  **An**
60. Three bells ring at intervals of 6, 12 and 18 minutes. If all the three bells rang at 6 a.m., when will they ring together again? (2023)  **U**
61. Find HCF and LCM of 404 and 96 and verify that $\text{HCF} \times \text{LCM} = \text{Product of the two given numbers}$. (2018)  **Ap**
62. An army contingent of 678 soldiers is to march behind an army band of 36 members in a Republic Day parade. The two groups are to march in the same number of columns. What is the maximum number of columns they can march? (Board Term I, 2017)

CBQ (4 marks)

63. **Case Study:** Teaching Mathematics through activities is a powerful approach that enhances students' understanding and engagement. Keeping this in mind, Ms. Mukta planned a prime number game for class 5 students. She announces the number 2 in her class and asked the first student to multiply it by a prime number and then pass it to second student. Second student also multiplied it by a prime number and passed it to third student. In this way by multiplying to a prime number, the last student got 173250. Now, Mukta asked some questions as given below to the students :
- What is the least prime number used by students?
 - (a) How many students are in the class?
- OR**
- (b) What is the highest prime number used by students?
 - (iii) Which prime number has been used maximum times? (2024)  **An**
64. **Case Study :** February 14 is celebrated as International Book Giving Day and many countries in the world celebrate this day. Some people in India also started celebrating this day and donated the following number of books of various subjects to a public library :
History = 96, Science = 240, Mathematics = 336.
These books have to be arranged in minimum number

of stacks such that each stack contains books of only one subject and the number of books on each stack is the same.

Based on the above information, answer the following questions :

- (i) How many books are arranged in each stack ?
- (ii) How many stacks are used to arrange all the Mathematics books ?
- (iii) (a) Determine the total number of stacks that will be used for arranging all the books.

OR

- (iii) (b) If the thickness of each book of History, Science and Mathematics is 1.8 cm, 2.2 cm and 2.5 cm respectively, then find the height of each stack of History, Science and Mathematics books.

(2023C) CFQ

LA (5 marks)

65. A sweet shopkeeper prepares 396 gulab jamuns and 342 ras-gullas. He packs them into containers. Each container consists of either gulab jamun or ras-gullas but have equal number of pieces. Find the number of pieces he should put in each box so that number of boxes are least. (Board Term I, 2017)
66. Find the largest possible positive integer that divides 125, 162 and 259 leaving remainder 5, 6 and 7 respectively. (Board Term I, 2017) 

1.3 Revisiting Irrational Numbers [Page 6-9]

MCQ

67. $\sqrt{0.4}$ is a/an
 - (a) natural number
 - (b) integer
 - (c) rational number
 - (d) irrational number (2025) 
68. Which of the following is a rational number between $\sqrt{3}$ and $\sqrt{5}$?
 - (a) 1.4142387954012....
 - (b) 2.326
 - (c) π
 - (d) 1.857142 (2025) 
69. A pair of irrational numbers whose product is a rational number is :
 - (a) $(\sqrt{16}, \sqrt{4})$
 - (b) $(\sqrt{5}, \sqrt{2})$
 - (c) $(\sqrt{3}, \sqrt{27})$
 - (d) $(\sqrt{36}, \sqrt{2})$ (2024) 

SA I (2 marks)

70. Prove that $4 - 2\sqrt{5}$ is an irrational number given that $\sqrt{5}$ is irrational. (2026) 
71. Prove that $2 - 5\sqrt{3}$ is an irrational number given that $\sqrt{3}$ is irrational. (2026) 

72. Prove that $7 - 3\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number. (2024C) 

73. Prove that $5 - 2\sqrt{3}$ is an irrational number.  It is given that $\sqrt{3}$ is an irrational number. (2024) 

74. If $\sqrt{2}$ is given as an irrational number, then prove that $(5 - 2\sqrt{2})$ is an irrational number. (2023C) 

75. Prove that $2 + \sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. (2023) 

76. Show that $5 + 2\sqrt{7}$ is an irrational number, where $\sqrt{7}$ is given to be an irrational number. (2020) 

77. Show that $\frac{3 + \sqrt{7}}{5}$ is an irrational number, given that $\sqrt{7}$ is irrational. (2019C) 

78. Given that $\sqrt{2}$ is irrational, prove that $(5 + 3\sqrt{2})$ is an irrational number. (2018)

79. How many irrational numbers lie between $\sqrt{2}$ and $\sqrt{3}$? Write any two of them. (Board Term I, 2017)

SA II (3 marks)

80. Prove that $\sqrt{3}$ is an irrational number.   (2026, 2024, 2023) 

81. Prove that $\sqrt{5}$ is an irrational number.    (2026, 2025, 2024, 2023, NCERT, AI 2019)

82. Prove that $\left(5\sqrt{3} + \frac{2}{3}\right)$ is an irrational number given that $\sqrt{3}$ is an irrational number.  (2025) 

83. Prove that $\left(4\sqrt{2} + \frac{5}{3}\right)$ is an irrational number given that $\sqrt{2}$ is an irrational number.  (2025) 

84. Prove that $(\sqrt{2} + \sqrt{3})^2$ is an irrational number, given that $\sqrt{6}$ is an irrational number. (2024) 

85. Prove that $\sqrt{2}$ is an irrational number. (2020 C, NCERT, Delhi 2019)

86. Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. (2019) 

LA (5 marks)

87. Define irrational number and prove that $3 + 2\sqrt{5}$ is an irrational number. (NCERT, Board Term I, 2017) 

Detailed SOLUTIONS

PREVIOUS YEARS' CBSE BOARD QUESTIONS

1. (c) : We have, $(3 \times 11 \times 13 + 3)$
 $= 3 \times (11 \times 13 + 1) = 3 \times (143 + 1)$
 $= 3 \times 144$, which is an even composite number.
2. (d) : The natural number 1 is neither prime nor composite.

Concept Applied

- ⇒ A composite number is a natural number or a positive integer which has more than two factors.

3. (a) : The natural number 2 is a prime number as its only divisors are 1 and 2.

4. (c) : We have,
 $(-1)^n + (-1)^8 = 0 \Rightarrow (-1)^n + 1 = 0 \Rightarrow (-1)^n = -1$
 On comparing the powers, we get
 $n = 1, 3, 5, \dots$
 $\therefore$ Hence, n is any odd number.

5. (c) : Total number of factors of a prime number is 2 i.e., 1 and number itself.

6. (a) : We have, $36m^2 = 2 \times 2 \times 3 \times 3 \times m \times m$
 and $18m = 2 \times 3 \times 3 \times m$
 $\therefore$ H.C.F. $(36m^2, 18m) = 2 \times 3 \times 3 \times m = 18m$
 So, both assertion (A) and reason (R) are true and reason (R) is the correct explanation of the assertion (A).

7. (c) : To distribute the books equally among students of both sections, the total number of books must be divisible by both 28 and 30.
 So, we find the LCM of 28 and 30.

$$28 = 2^2 \times 7$$

$$30 = 2 \times 3 \times 5$$

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Therefore, the minimum number of books required is 420.

8. (b)

9. (a) : Prime factorisation of 960 and 240 are
 $960 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5$
 $240 = 2 \times 2 \times 2 \times 2 \times 3 \times 5$
 Hence, $\text{LCM}(960, 240) = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5 = 960$

Expert Tips

- ⇒ The LCM is the value of two numbers that is evenly divisible by the given numbers.

10. (a) : Prime factorization of 960 and 432 are
 $960 = 2^6 \times 3 \times 5$
 $432 = 2^4 \times 3^3$
 $\therefore$ HCF $(960, 432) = 2^4 \times 3 = 16 \times 3 = 48$

11. (b) : The power of 6 are $6^1 = 6$; $6^2 = 36$; $6^3 = 216$; $6^4 = 1296$

The units digit for any natural power of 6 is always 6.

12. (c) : The total number of factors of the square of a prime number (p^2) is 3 i.e., 1, p , p^2 .
 For example : $2^2 = 4$, factors of 4 are 1, 2, 4.
 $5^2 = 25$, factors of 25 are 1, 5, 25.

13. (c) : $8^n = (2 \times 2 \times 2)^n$
 Since, factors of 8^n do not contain 5 in it.
 So, 8^n can't ends with 0.
 $\Rightarrow$ 0 can't be the unit digit of 8^n .

14. (a) : LCM $(4, 6, 8) = x = 24$
 LCM $(3, 5, 7) = y = 105$
 LCM $(x, y) = p = \text{LCM}(24, 105) = 840$
 $\therefore p = 840 = 35 \times 24 = 35x$

15. (c) : $98 = 2 \times 7^2$, $28 = 2^2 \times 7$
 $\therefore \text{LCM}(98, 28) = 2^2 \times 7^2 = 196$
 $\Rightarrow n = 196$
 HCF $(98, 28) = 2 \times 7 = 14$
 $\Rightarrow m = 14$
 $\therefore n - 7m = 196 - 7 \times 14 = 196 - 98 = 98$

16. (a) : We have to find the H.C.F of $70 - 5 = 65$ and $125 - 8 = 117$.
 $\therefore 65 = 5 \times 13$, $117 = 3^2 \times 13$
 $\Rightarrow \text{HCF}(65, 117) = 13$
 $\therefore$ Required greatest number = 13.

17. (d) : Clearly, reason is true.
 For any two primes p and q .
 HCF $(p, q) = 1$
 LCM $(p, q) = p \times q$
 $\therefore$ Assertion is false.

18. (c) : Given, m, n are two positive integers, $m = p^5 q^2$ and $n = p^3 q^4$, where p and q are prime numbers.
 LCM of m and n is the product of all the prime factors raised to their highest exponents.
 $\therefore \text{LCM}(m, n) = p^5 q^4$

19. (d) : We have,
 $p = 18a^2 b^4 = 2 \times 3 \times 3 \times a \times a \times b \times b \times b \times b$
 $q = 20a^3 b^2 = 2 \times 2 \times 5 \times a \times a \times a \times b \times b$
 $\therefore \text{LCM}(p, q) = 2 \times 2 \times 3 \times 3 \times 5 \times a \times a \times a \times b \times b \times b \times b$
 $= 180a^3 b^4$

20. (c) : We have, $a = 2^2 \times 3^x$, $b = 2^2 \times 3 \times 5$, $c = 2^2 \times 3 \times 7$
 LCM $(a, b, c) = 2^2 \times 3^x \times 5 \times 7$
 $\Rightarrow 2^2 \times 3^x \times 5 \times 7 = 3780$ ($\therefore$ Given)
 $\Rightarrow 140 \times 3^x = 3780 \Rightarrow 3^x = 27 \Rightarrow 3^x = 3^3$
 $\therefore x = 3$

21. (a) : $\text{HCF}(2550, 6600) = 40$

Now, $\text{LCM}(2520, 6600) = 252 \times k$

$\text{HCF} \times \text{LCM} = \text{Product of numbers}$

$\therefore 40 \times 252 \times k = 2520 \times 6600 = 16632000 = 40 \times 252 \times k$

$\Rightarrow k = 1650$

Concept Applied

- Product of two numbers is equal to the product of their HCF and LCM.

Note : There is an error in question as $\text{HCF}(2520, 6600) = 120$.

22. (d) : Smallest prime number = 2

Smallest odd composite number = 9

$\therefore \text{LCM}(2, 9) = 18$

23. (c) : We have to find the greatest number which divides 281 and 1249, leaving remainder 5 and 7 respectively. This means that $281 - 5 = 276$ and $1249 - 7 = 1242$ are divisible by the required number.

$\therefore$ Required number is HCF of 276 and 1242.

Prime factorisation of 276 and 1242 are

$276 = 2 \times 2 \times 3 \times 23$

$1242 = 2 \times 3 \times 3 \times 3 \times 23$

$\therefore \text{HCF}(276, 1242) = 2 \times 3 \times 23 = 138$

24. (d) : Prime factorisation of 28, 44 and 132 are

$28 = 2 \times 2 \times 7$

$44 = 2 \times 2 \times 11$

$132 = 2 \times 2 \times 3 \times 11$

$\therefore \text{LCM}(28, 44, 132) = 2 \times 2 \times 7 \times 3 \times 11 = 924$

25. (b) : Any number that ends with the digit 0 must have a prime factorisation that includes atleast one factor of 2 and atleast one factor of 5.

Since, 5^n has only one prime factor of 5, so it cannot end with the digit 0.

$\therefore$ Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

26. (c) : Since, p is a multiple of q . So, $p = nq$ where n is any natural number.

$\text{HCF of } p \text{ and } q = \text{HCF of } nq \text{ and } q = q$.

27. (a) : Least composite number = 4

Least prime number = 2

$\therefore \text{HCF} = 2, \text{LCM} = 4$

$\therefore$ Required ratio = $\frac{2}{4}$ i.e., 1:2

28. (b) : We know that,

$\text{HCF} \times \text{LCM} = \text{Product of two numbers}$

$\Rightarrow 13 \times \text{LCM} = 39 \times 91$

$\Rightarrow \text{LCM} = \frac{39 \times 91}{13} = 273$

29. (a) : Given, $\text{HCF} = 12$

Let two numbers be $12a$ and $12b$

So, $12a \times 12b = 6336 \Rightarrow ab = 44$

We can write 44 as product of two numbers in these ways :

$ab = 1 \times 44 = 2 \times 22 = 4 \times 11$

Here, we will take $a = 1$ and $b = 44$; $a = 4$ and $b = 11$.

We do not take $ab = 2 \times 22$ because 2 and 22 are not co-prime to each other.

For $a = 1$ and $b = 44$, 1^{st} no. = $12a = 12$, 2^{nd} no. = $12b = 528$

For $a = 4$ and $b = 11$, 1^{st} no. = $12a = 48$, 2^{nd} no. = $12b = 132$

Hence, we get two pairs of numbers, (12, 528) and (48, 132).

Expert Tips

- Don't mixup the concept of prime and co-prime numbers.
- Remember the difference between prime and co-prime numbers.
- A prime number has exactly two distinct positive divisors 1 and itself, whereas co-prime numbers are pair of numbers that share no common positive divisors other than 1.

30. (d) : For $n = 1, 2, 3, 4 \dots$, $(12)^n$ cannot end with 0.

31. (b) : We have,

5	385
7	77
11	11
	1

$\therefore$ Prime factorisation of 385 = $5 \times 7 \times 11$

32. (c) : We have, $12 = 2 \times 2 \times 3 = 2^2 \times 3$

$21 = 3 \times 7$

$15 = 3 \times 5$

$\therefore \text{HCF}(12, 21, 15) = 3$ and

$\text{LCM}(12, 21, 15) = 2^2 \times 3 \times 5 \times 7 = 420$

33. Let the other number be x .

As, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

$\Rightarrow 13 \times 182 = 26 \times x \Rightarrow x = \frac{13 \times 182}{26} = 91$

Hence, other number is 91.

34. Let a and b be two numbers such that

$\text{LCM}(a, b) = 9 \times \text{HCF}(a, b)$

and $\text{LCM}(a, b) + \text{HCF}(a, b) = 500$

Using (i) in (ii), we get

$9\text{HCF}(a, b) + \text{HCF}(a, b) = 500$

$\Rightarrow 10\text{HCF}(a, b) = 500$

$\Rightarrow \text{HCF}(a, b) = 50$

35. Since, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

$\therefore \text{HCF}(336, 54) \times \text{LCM}(336, 54) = 336 \times 54$

$\Rightarrow 6 \times \text{LCM}(336, 54) = 18144$

$\Rightarrow \text{LCM}(336, 54) = \frac{18144}{6} = 3024$

36. We know that, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

$\Rightarrow 5 \times 200 = ab \Rightarrow ab = 1000$

37. Smallest prime number = 2

Smallest composite number = 4

$$\therefore \text{HCF}(2, 4) = 2$$

Concept Applied

- A composite number is a natural number or a positive integer which has more than two factors.

38. The prime factors of $6^n = (2 \times 3)^n = 2^n \times 3^n$.

Therefore prime factorisation of 6^n does not contain prime factor 5. Hence, 6^n can never end with the digit 0 for any natural number n .

39. Let the other number be x .

As, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

$$\therefore 27 \times 162 = 54x \Rightarrow x = \frac{27 \times 162}{54} = 81$$

Hence, other number is 81.

40. We have, 4 m 20 cm = 420 cm

and 5 m 4 cm = 504 cm

Now, $420 = 2 \times 2 \times 3 \times 5 \times 7$

$$504 = 2 \times 2 \times 2 \times 3 \times 3 \times 7$$

So, $\text{HCF}(420, 504) = 2 \times 2 \times 3 \times 7 = 84$ cm

Therefore, a plank of length 84 cm can be used to measure the lengths 4 m 20 cm and 5 m 4 cm exactly. (1)

41.	3	51, 85
	5	17, 85
	17	17, 17
		1, 1

$$\therefore \text{L.C.M.}(51, 85) = 3 \times 5 \times 17$$

$$\Rightarrow 6z - 9 = 3 \times 5 \times 17$$

$$\Rightarrow 2z - 3 = 5 \times 17$$

$$\Rightarrow 2z - 3 = 85$$

$$\Rightarrow 2z = 88$$

$$\Rightarrow z = \frac{88}{2} = 44$$

42. The smallest number divisible by both 644 and 462 is LCM of 644 and 462.

$$644 = 2 \times 2 \times 7 \times 23$$

$$462 = 2 \times 3 \times 7 \times 11$$

$$\therefore \text{LCM}(644, 462) = 2 \times 2 \times 3 \times 7 \times 11 \times 23 = 21252 \quad (2)$$

43. Let two numbers be $4x$ and $5x$.

$$\text{HCF}(4x, 5x) = 11$$

$$\therefore \text{Numbers are } 4 \times 11 = 44 \text{ and } 5 \times 11 = 55 \quad (1)$$

Since, product of two numbers = $\text{HCF} \times \text{LCM}$

$$\Rightarrow 44 \times 55 = 11 \times \text{LCM}$$

$$\Rightarrow \frac{44 \times 55}{11} = \text{LCM} \Rightarrow \text{LCM} = 220 \quad (1)$$

44. We have, $5 \times 11 \times 17 + 3 \times 11 = 11(5 \times 17 + 3)$
 $= 11 \times (85 + 3) = 11 \times 88$

Now, 11×88 is divisible by 2, 4, 8 and 11. (1)

Since, it has more than 2 factors. So, the given number is a composite number. (1)

45. The prime factors of $6^n = (2 \times 3)^n = 2^n \times 3^n$ i.e., 2^n and 3^n , where n is any natural number. (1/2)

As, the prime factorisation of 6^n does not contain 5. (1/2)

$\therefore 6^n$ can never end with the digit 0 for any natural number n . (1)

46. Given, least number which when divided by 12, 16 and 24 leaves remainder 7 in each case

$$\therefore \text{Least number} = \text{LCM}(12, 16, 24) + 7 = 48 + 7 = 55 \quad (2)$$

47. Let the two numbers be $2x$ and $3x$.

$$\text{LCM of } 2x \text{ and } 3x = 6x, \text{HCF}(2x, 3x) = x \quad (1/2)$$

$$\text{Now, } 6x = 180 \quad [\text{Given}]$$

$$\Rightarrow x = \frac{180}{6} = 30 \quad (1/2)$$

$$\therefore \text{HCF}(2x, 3x) = x = 30 \quad (1)$$

48. We have, $2 \times 3 \times 5 + 5$ and $5 \times 7 \times 11 + 7 \times 5$

We can write these numbers as:

$$2 \times 3 \times 5 + 5 = 5(2 \times 3 + 1) = 1 \times 5 \times 7$$

$$\text{and } 5 \times 7 \times 11 + 7 \times 5 = 5 \times 7(11 + 1)$$

$$= 5 \times 7 \times 12 = 1 \times 5 \times 7 \times 12 \quad (1)$$

Since, on simplifying, we find that both the numbers have more than two factors. So, these are composite numbers. (1)

49. Since, $\text{HCF}(65, 117) = 13$

$$\text{Given } \text{HCF}(65, 117) = 65n - 117 \quad (1)$$

$$\Rightarrow 13 = 65n - 117 \Rightarrow 65n = 13 + 117 \Rightarrow n = 2 \quad (1)$$

50. Prime factorisation of 612 and 1314 are

$$612 = 2 \times 2 \times 3 \times 3 \times 17$$

$$1314 = 2 \times 3 \times 3 \times 73 \quad (1)$$

$$\therefore \text{HCF}(612, 1314) = 2 \times 3 \times 3 = 18 \quad (1)$$

$$51. 5050 = 2 \times 5 \times 5 \times 101 = 2 \times 5^2 \times 101$$

Yes, it is unique. (2)

52. Prime factorisation of 231 and 396 are

$$231 = 3 \times 7 \times 11$$

$$396 = 2 \times 2 \times 3 \times 3 \times 11 \quad (1)$$

$$\text{HCF}(231, 396) = 3 \times 11 = 33 \neq 1$$

Hence, the two numbers are not co-prime. (1)

Concept Applied

- Every natural number greater than 1 can be written as a product of prime numbers and that up to rearrangement of the factors, the product is unique.

53. First we find the H.C.F of 156 and 216 for maximum size of squares.

$$156 = 2 \times 2 \times 3 \times 13$$

$$216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3$$

$$\therefore \text{HCF}(156, 216) = 2 \times 2 \times 3 = 12$$

So, 12 cm is the side of square. (2)

$$\therefore \text{Number of squares formed} = \frac{156 \times 216}{12 \times 12} = 13 \times 18 = 234 \quad (1)$$

54. We first find the L.C.M. of 15, 18 and 20.

2	15, 18, 20
2	15, 9, 10
3	15, 9, 5
3	5, 3, 5
5	5, 1, 5
	1, 1, 1

$\therefore$ L.C.M. (15, 18 and 20) = $2 \times 2 \times 3 \times 3 \times 5 = 180$

$\therefore$ After 180 days all of them will start making a new sweater again. (1½)

By that time, number of sweaters completed by Ranjita

$$= \frac{180}{15} = 12$$

By that time, number of sweaters completed by Neha

$$= \frac{180}{18} = 10$$

By that time, number of sweaters completed by Salma

$$= \frac{180}{20} = 9 \quad (1½)$$

55. Number of Physics books = 144

Number of Chemistry books = 180

Number of Mathematics books = 192 (1/2)

Since, $180 = 2 \times 2 \times 3 \times 3 \times 5$

$$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$$

$$192 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3$$

$\therefore$ HCF(180, 144, 192) = $2 \times 2 \times 3 = 12$ (1)

$\therefore$ Number of stacks of Physics books = $\frac{144}{12} = 12$ (1/2)

Number of stacks of Chemistry books = $\frac{180}{12} = 15$ (1/2)

Number of stacks of Mathematics books = $\frac{192}{12} = 16$ (1/2)

56. Given : p, q, r be distinct prime numbers.

We have, $pqr + q = q(pr + 1)$

Here, q is a prime number and $pr + 1 > 1$,

Thus, $pqr + q$ has factors 1, q , $(pr + 1)$ and $q(pr + 1)$.

Hence, $pqr + q$ is a composite number. ... (i) (1)

(i) Take $p = 3, q = 5$ and $r = 7$, we have

$$pqr + 1 = 3 \times 5 \times 7 + 1 = 105 + 1 = 106$$

Thus, $pqr + 1$ is a composite number for $p = 3, q = 5$ and $r = 7$. (1)

(ii) Take $p = 2, q = 3$ and $r = 5$, we have

$$pqr + 1 = 2 \times 3 \times 5 + 1 = 30 + 1 = 31$$

Thus, $pqr + 1$ is a prime number for $p = 2, q = 3$ and $r = 5$. (1)

57. Number of Maths teacher = $42 = 2 \times 3 \times 7$

Number of Physics teacher = $56 = 2 \times 2 \times 2 \times 7$

Number of Chemistry teacher = $70 = 2 \times 5 \times 7$

Maximum number of teachers of same subject that can sit

$$\text{at a table} = \text{HCF}(42, 56, 70) = 2 \times 7 = 14 \quad (1½)$$

$$\text{Number of tables required for Maths teacher} = \frac{42}{14} = 3$$

$$\text{Number of tables required for Physics teacher} = \frac{56}{14} = 4$$

$$\text{Number of tables required for Chemistry teacher} = \frac{70}{14} = 5$$

$$\therefore \text{Minimum number of tables required} = 3 + 4 + 5 = 12 \quad (1½)$$

58. The number of rooms will be minimum, if the number of teachers seated in each room is maximum.

So, we have to find the HCF of 48, 80 and 144

As, prime factorisation of;

$$48 = 2 \times 2 \times 2 \times 2 \times 3,$$

$$80 = 2 \times 2 \times 2 \times 2 \times 5 \text{ and } 144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3$$

$$\therefore \text{H.C.F. of } (48, 80, 144) = 2 \times 2 \times 2 \times 2 = 16 \quad (1½)$$

$$\text{So, number of rooms required} = \frac{48}{16} + \frac{80}{16} + \frac{144}{16} = 3 + 5 + 9 = 17 \quad (1½)$$

Expert Tips

➔ HCF (Highest Common Factor) is also called GCD (Greatest Common Divisor)

59. We have,

$$18180 = 2 \times 2 \times 3 \times 3 \times 5 \times 101$$

$$\text{and } 7575 = 3 \times 5 \times 5 \times 101 \quad (1)$$

$\therefore$ LCM of 18180 and 7575

$$= 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 101$$

$$= 90900$$

$$\text{Also, HCF} = 3 \times 5 \times 101 = 1515$$

60. Given that the three bells ring at intervals of 6, 12 and 18 minutes.

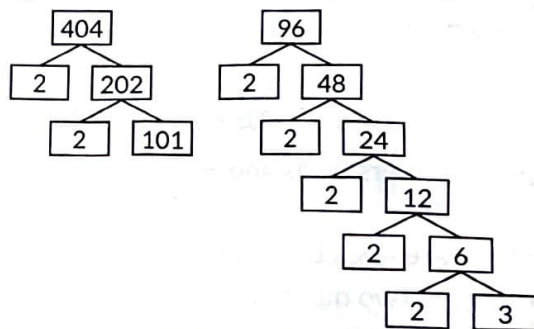
Let us find LCM of 6, 12 and 18. (1/2)

$$6 = 2 \times 3, 12 = 2 \times 2 \times 3 \text{ and } 18 = 2 \times 3 \times 3 \quad (1/2)$$

$$\therefore \text{LCM of } 6, 12 \text{ and } 18 = 2 \times 2 \times 3 \times 3 = 36 \quad (1)$$

So, after 36 minutes the bells will ring together again i.e., at 6 : 36 a.m. (1)

61. Using the factor tree method, we have



$$\Rightarrow 404 = 2 \times 2 \times 101 \text{ and } 96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$$

$$\therefore \text{HCF of } 404 \text{ and } 96 = 2 \times 2 = 4 \quad (1)$$

$$\text{LCM of } 404 \text{ and } 96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 101 = 9696$$

$$\text{Also } 404 \times 96 = 38784 \quad (1)$$

$$\text{LCM} \times \text{HCF} = 9696 \times 4 = 38784$$

$$\text{Thus, HCF} \times \text{LCM} = \text{Product of two numbers.} \quad (1)$$

62. Number of soldiers in an army contingent = 678
 $= 2 \times 3 \times 113$
 Number of members in an army band = $36 = 2 \times 2 \times 3 \times 3$
 (1)
 The maximum number of columns such that two groups can march in same number of columns is HCF of 678 and 36.
 $\text{HCF}(678, 36) = 2 \times 3 = 6$
 So, the maximum number of columns they can march is 6.
 (2)

Expert Tips

- ⇒ HCF (Highest Common Factor) is also called GCD (Greatest Common Divisor)

63.
$$\begin{array}{r|l} 2 & 173250 \\ 3 & 86625 \\ 3 & 28875 \\ 5 & 9625 \\ 5 & 1925 \\ 5 & 385 \\ 7 & 77 \\ 11 & 11 \\ & 1 \end{array}$$

- ∴ $173250 = 2 \times 3 \times 3 \times 5 \times 5 \times 5 \times 7 \times 11$
 (i) Least prime number used by students = 3 (1)
 (ii) (a) Number of students = Number of prime factors other than 2
 $= 7$
 OR
 (ii) (b) Highest prime number used by students = 11 (2)
 (iii) 5 is used maximum number of times as
 $173250 = 2 \times 3^2 \times 5^3 \times 7 \times 11$ (1)

64. Given, Number of History books = 96
 Number of Science books = 240
 Number of Mathematics books = 336
 Now, $96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$
 $240 = 2 \times 2 \times 2 \times 2 \times 3 \times 5$
 $336 = 2 \times 2 \times 2 \times 2 \times 3 \times 7$
 $\text{H.C.F.}(96, 240, 336) = 2 \times 2 \times 2 \times 2 \times 3 = 48$
 (i) Since, H.C.F. (96, 240, 336) = 48
 48 books can be arranged in each stack. (1)
 (ii) Since, number of Mathematics books = 336
 Number of stack for 48 books = 1

- ∴ Number of stacks for 336 books = $\frac{1}{48} \times 336 = 7$
 7 stacks are used to arrange all the Mathematics books. (1)

- (iii) (a) Since, number of stacks used to arrange,

History books = $\frac{96}{48} = 2$,

Science books = $\frac{240}{48} = 5$ and

Mathematics books = $\frac{336}{48} = 7$ (1½)

Total number of stacks used to arrange all the books
 $= 2 + 5 + 7 = 14$ (1/2)

OR

- (iii) (b) Given, thickness of a History book = 1.8 cm
 Thickness of a Science book = 2.2 cm
 Thickness of a Mathematics book = 2.5 cm (1)
 So, height of stack of History book = $1.8 \times 48 = 86.4$ cm
 Height of stack of Science book = $2.2 \times 48 = 105.6$ cm
 Height of stack of Mathematics book = $2.5 \times 48 = 120$ cm (1)
65. Number of gulab jamuns = $396 = 2 \times 2 \times 3 \times 3 \times 11$ (1)
 Number of ras-gullas = $342 = 2 \times 3 \times 3 \times 19$ (1)
 $\text{HCF}(396, 342) = 2 \times 3 \times 3 = 18$ (2)
 So, shopkeeper will put 18 sweets in each box such that number of boxes are least. (1)

66. It is given that the required number when divides 125, 162, 259 leaves the remainder 5, 6, 7 respectively. This means that $125 - 5 = 120$, $162 - 6 = 156$, $259 - 7 = 252$ are divisible by the required number. (2)
 The required number is HCF of all these numbers. (1)
 The prime factorisation of 120, 156, 252 are
 $120 = 2 \times 2 \times 2 \times 3 \times 5$
 $156 = 2 \times 2 \times 3 \times 13$; $252 = 2 \times 2 \times 3 \times 3 \times 7$
 $\text{HCF}(120, 156, 252) = 2 \times 2 \times 3 = 12$
 Hence, the required number is 12. (2)

Expert Tips

- ⇒ The LCM is the value of two numbers that is evenly divisible by the given numbers.

67. (d) : $\sqrt{0.4} = \sqrt{\frac{4}{10}} = \frac{2}{\sqrt{10}}$

∴ 2 is a rational number and $\sqrt{10}$ is an irrational number.

∴ $\sqrt{0.4}$ is an irrational number.

68. (d) : $\sqrt{3}$ is approximately equals to 1.7320.

$\sqrt{5}$ is approximately equals to 2.2360.

Here, 1.857142 lies between $\sqrt{3}$ and $\sqrt{5}$.

69. (c) : $\sqrt{3 \times 27} = \sqrt{3 \times 3 \times 3 \times 3} = 3 \times 3 = 9$, which is a rational number.

70. Let us assume $4 - 2\sqrt{5}$ be a rational number.

Then, $4 - 2\sqrt{5} = \frac{a}{b}$, where $b \neq 0$ and $a, b \in \mathbb{R}$

$\Rightarrow 2\sqrt{5} = 4 - \frac{a}{b} \Rightarrow \sqrt{5} = \frac{1}{2} \left(4 - \frac{a}{b} \right)$, which is a rational number. (1)

$\Rightarrow \sqrt{5}$ is a rational number which contradicts that $\sqrt{5}$ is an irrational number. So, our assumption is wrong.

Hence, $4 - 2\sqrt{5}$ is an irrational number. (1)

Concept Applied

➤ Irrational number is a number which can not be written in the form of p/q , where p and q are integers and $q \neq 0$.

71. Suppose $2 - 5\sqrt{3}$ is a rational number.
 $\therefore$ We can find two integers a, b ($b \neq 0$) such that

$$2 - 5\sqrt{3} = \frac{a}{b}, \text{ where } a \text{ and } b \text{ are co-prime integers.}$$

$$\Rightarrow 5\sqrt{3} = 2 - \frac{a}{b} \Rightarrow \sqrt{3} = \frac{1}{5} \left[2 - \frac{a}{b} \right]$$

$$\Rightarrow \sqrt{3} \text{ is a rational number.}$$

$$[\because a, b \text{ are integers, so } \frac{1}{5} \left[2 - \frac{a}{b} \right] \text{ is a rational number.}] \quad (1)$$

But this contradicts the fact that $\sqrt{3}$ is an irrational number.

Hence, our assumption is wrong.

Thus, $2 - 5\sqrt{3}$ is an irrational number. (1)

72. Let $7 - 3\sqrt{5}$ be a rational number.

$$\Rightarrow 7 - 3\sqrt{5} = \frac{a}{b} \quad [\text{Where } a, b \text{ are coprime and } b \neq 0]$$

$$\Rightarrow 7 - \frac{a}{b} = 3\sqrt{5} \Rightarrow \frac{7b - a}{b} = 3\sqrt{5} \Rightarrow \frac{7b - a}{3b} = \sqrt{5} \quad (1)$$

$\Rightarrow \sqrt{5}$ is a rational, but $\sqrt{5}$ is an irrational number. Thus, our assumption is wrong.

So, $7 - 3\sqrt{5}$ is an irrational number. (1)

73. It is given that, $\sqrt{3}$ is an irrational number.

We have to prove that $5 - 2\sqrt{3}$ is an irrational number.

Let us assume $5 - 2\sqrt{3}$ be a rational number.

$$\text{Then, } 5 - 2\sqrt{3} = \frac{a}{b}, \text{ where } b \neq 0 \text{ and } a, b \in \mathbb{R}$$

$$\Rightarrow 2\sqrt{3} = 5 - \frac{a}{b}$$

$$\Rightarrow \sqrt{3} = \left(\frac{5}{2} - \frac{a}{2b} \right), \text{ which is a rational number.} \quad (1)$$

$$\Rightarrow \sqrt{3} \text{ is a rational number.}$$

But $\sqrt{3}$ is an irrational number. So, our assumption is wrong.

Hence, $5 - 2\sqrt{3}$ is an irrational number. (1)

Concept Applied

➤ Irrational number is a number which can not be written in the form of p/q , where p and q are integers and $q \neq 0$.

74. Suppose that $5 - 2\sqrt{2}$ is a rational number.

$\therefore$ We can find two integers a, b ($b \neq 0$) such that

$$5 - 2\sqrt{2} = \frac{a}{b}, \text{ where } a \text{ and } b \text{ are co-prime integers.}$$

$$\Rightarrow 2\sqrt{2} = 5 - \frac{a}{b} \Rightarrow \sqrt{2} = \frac{1}{2} \left(5 - \frac{a}{b} \right)$$

$$\Rightarrow \sqrt{2} \text{ is a rational number} \quad (1)$$

$$(\because a, b \text{ are integers, } \therefore \frac{1}{2} \left(5 - \frac{a}{b} \right) \text{ is a rational number.})$$

But this contradicts the fact that $\sqrt{2}$ is an irrational number.

Hence, our supposition is wrong.

Thus, $5 - 2\sqrt{2}$ is an irrational number. (1)

75. Let $(2 + \sqrt{3})$ is a rational number.

$$\therefore 2 + \sqrt{3} = \frac{p}{q}, \text{ where } p \text{ and } q \text{ are integers, } q \neq 0 \text{ and } p \text{ and } q \text{ are co-prime numbers.}$$

$$\Rightarrow \sqrt{3} = \frac{p}{q} - 2 \Rightarrow \sqrt{3} = \frac{p - 2q}{q} \quad (1)$$

$$\text{Since, } \frac{p - 2q}{q} \text{ is a rational.}$$

$$\Rightarrow \sqrt{3} \text{ is rational but it is given that } \sqrt{3} \text{ is an irrational number.}$$

So, our assumption is wrong and hence $2 + \sqrt{3}$ is an irrational number. (1)

76. Suppose $5 + 2\sqrt{7}$ is a rational number.

$\therefore$ We can find two integers a, b ($b \neq 0$) such that

$$5 + 2\sqrt{7} = \frac{a}{b}, \text{ where } a \text{ and } b \text{ are co-prime.}$$

$$\Rightarrow 2\sqrt{7} = \frac{a}{b} - 5 \Rightarrow \sqrt{7} = \frac{1}{2} \left[\frac{a}{b} - 5 \right] \quad (1)$$

$$\Rightarrow \sqrt{7} \text{ is a rational number}$$

$$[\because a, b \text{ are integers, so } \frac{1}{2} \left(\frac{a}{b} - 5 \right) \text{ is a rational number}]$$

But this contradicts the fact that $\sqrt{7}$ is an irrational number. Hence, our assumption is wrong. Thus, $5 + 2\sqrt{7}$ is an irrational number. (1)

77. Suppose $\frac{3 + \sqrt{7}}{5}$ is rational number.

$\therefore$ We can find two integers p and q ($q \neq 0$) such that

$$\frac{3 + \sqrt{7}}{5} = \frac{p}{q}, \text{ where } p \text{ and } q \text{ are co-prime.}$$

$$\Rightarrow 3 + \sqrt{7} = \frac{5p}{q}$$

$$\Rightarrow \sqrt{7} = \frac{5p}{q} - 3 \Rightarrow \sqrt{7} \text{ is a rational number.} \quad (1)$$

$$[\because p \text{ and } q \text{ are integers, so } \frac{5p}{q} - 3 \text{ is a rational number}]$$

But this contradicts the fact that $\sqrt{7}$ is an irrational number.

Hence, our supposition is wrong.

Thus, $\frac{3 + \sqrt{7}}{5}$ is an irrational number. (1)

78. Let $(5 + 3\sqrt{2})$ is rational.

Then, $5 + 3\sqrt{2} = \frac{a}{b}$, where $a, b (\neq 0)$ are coprime numbers

$$\therefore 3\sqrt{2} = \frac{a}{b} - 5 \Rightarrow \sqrt{2} = \frac{a-5b}{3b}$$

$\Rightarrow \sqrt{2}$ is rational number.

(1)

$[\because a, b \text{ are integers } \therefore \frac{a-5b}{3b} \text{ is rational}]$

But this contradicts the fact that $\sqrt{2}$ is irrational.

Hence, $5 + 3\sqrt{2}$ is also an irrational number.

(1)

79. There are infinite irrational numbers between

$\sqrt{2}$ and $\sqrt{3}$. Two of them are $\sqrt{2.1}$ and $\sqrt{2.3}$.

(2)

80. Let us assume that $\sqrt{3}$ is a rational number.

Then $\sqrt{3} = \frac{a}{b}$; where a and $b (\neq 0)$ are co-prime positive integers.

Squaring on both sides, we get

$$3 = \frac{a^2}{b^2} \Rightarrow a^2 = 3b^2 \Rightarrow 3 \text{ divides } a^2$$

$\Rightarrow 3$ divides a

...(i) (1)

$\Rightarrow a = 3c$, where c is an integer

Again, squaring on both sides, we get

$$a^2 = 9c^2$$

$$\Rightarrow 3b^2 = 9c^2 \Rightarrow b^2 = 3c^2 \Rightarrow 3 \text{ divides } b^2$$

$\Rightarrow 3$ divides b

...(ii) (1)

From (i) and (ii), we get 3 divides both a and b .

$\Rightarrow a$ and b are not co-prime integers.

This contradicts the fact that a and b are co-primes.

Hence, $\sqrt{3}$ is an irrational number.

(1)

Concept Applied

Co-prime numbers are pairs of numbers that do not have any common factor other than 1.

81.

let us assume to the contrary that $\sqrt{5}$ is a rational no. of form $\frac{p}{q}$ where p and q are integers and co-prime and $q \neq 0$.

$\therefore \sqrt{5} = \frac{p}{q}$

squaring, $(\sqrt{5})^2 = (\frac{p}{q})^2$

$5 = \frac{p^2}{q^2}$

$q^2 = \frac{p^2}{5}$ (1)

Thus, 5 divides p^2
5 divides p as well. - (1)

let $p = 5a$

putting $p = 5a$ in eq (1)

$q^2 = \frac{(5a)^2}{5}$

$q^2 = 5a^2$

$q^2 = 5a^2$

Thus, 5 divides q^2
5 divides q as well. - (2)

From statement (1) and (2), it is clear that 5 is a common factor of p & q , but we assumed that p and q are co-prime! This contradiction has arisen due to our incorrect assumption that $\sqrt{5}$ is rational.

Hence proved, $\sqrt{5}$ is irrational.

[Topper's Answer, 2025]

82. It is given that, $\sqrt{3}$ is an irrational number.

We have to prove that $5\sqrt{3} + \frac{2}{3}$ is an irrational number.

Let us assume $5\sqrt{3} + \frac{2}{3}$ be a rational number.

Then, $5\sqrt{3} + \frac{2}{3} = \frac{a}{b}$, where $b \neq 0$ and a, b are co-prime integers.

$$\Rightarrow 5\sqrt{3} = \frac{a}{b} - \frac{2}{3}$$

(1)

$$\Rightarrow \sqrt{3} = \frac{1}{5} \left(\frac{a}{b} - \frac{2}{3} \right) = \frac{3a-2b}{15b}, \text{ which is a rational number.}$$

$\Rightarrow \sqrt{3}$ is a rational number.

(1)

But $\sqrt{3}$ is an irrational number. So, our assumption is wrong.

Hence, $5\sqrt{3} + \frac{2}{3}$ is an irrational number.

(1)

83. It is given that, $\sqrt{2}$ is an irrational number.

We have to prove that $\left(4\sqrt{2} + \frac{5}{3}\right)$ is an irrational number.

Let us assume $\left(4\sqrt{2} + \frac{5}{3}\right)$ be a rational number.

Then, $4\sqrt{2} + \frac{5}{3} = \frac{a}{b}$, where $b \neq 0$ and $a, b \in \mathbb{Z}$

$$\Rightarrow 4\sqrt{2} = \frac{a}{b} - \frac{5}{3}$$

(1)

$$\Rightarrow 4\sqrt{2} = \frac{3a-5b}{3b}$$

$$\Rightarrow \sqrt{2} = \left(\frac{3a-5b}{12b} \right), \text{ since } \frac{3a-5b}{12b} \text{ is a rational number.}$$

$\Rightarrow \sqrt{2}$ is a rational number.

(1)

But $\sqrt{2}$ is an irrational number. So, our assumption is wrong.

Hence, $\left(4\sqrt{2} + \frac{5}{3}\right)$ is an irrational number.

(1)

84. We have, $(\sqrt{2} + \sqrt{3})^2 = 2 + 3 + 2\sqrt{6} = 5 + 2\sqrt{6}$

Let $5 + 2\sqrt{6}$ be a rational number.

$\therefore 5 + 2\sqrt{6} = \frac{a}{b}$, where a and b are coprime numbers and

$b \neq 0$.

$$\Rightarrow 2\sqrt{6} = \frac{a}{b} - 5 \Rightarrow \sqrt{6} = \frac{1}{2} \left(\frac{a}{b} - 5 \right) = \frac{a}{2b} - \frac{5}{2} \quad \dots(i) \quad (1)$$

Since, $\frac{a}{b}$ is a rational number

$\therefore \frac{a}{2b}$ is also a rational number. (1)

Hence, $\frac{a}{2b} - \frac{5}{2}$ is a rational number.

From (i), $\sqrt{6}$ is an irrational number but $\frac{a}{2b} - \frac{5}{2}$ is a rational number, which is a contradiction.

Therefore, $(\sqrt{2} + \sqrt{3})^2$ is an irrational number.

Hence, proved. (1)

85. Let us assume $\sqrt{2}$ be a rational number.

Then, $\sqrt{2} = \frac{p}{q}$, where p, q ($q \neq 0$) are integers and co-prime.

On squaring both sides, we get

$$2 = \frac{p^2}{q^2} \Rightarrow p^2 = 2q^2 \quad \dots(i)$$

$$\Rightarrow 2 \text{ divides } p^2 \Rightarrow 2 \text{ divides } p \quad \dots(ii) \quad (1)$$

So, $p = 2a$, where a is some integer.

Again squaring on both sides, we get

$$p^2 = 4a^2 \Rightarrow 2q^2 = 4a^2 \quad \text{(using (i))}$$

$$\Rightarrow q^2 = 2a^2$$

$$\Rightarrow 2 \text{ divides } q^2 \Rightarrow 2 \text{ divides } q \quad \dots(iii) \quad (1)$$

From (ii) and (iii), we get

2 divides both p and q .

$\therefore p$ and q are not co-prime integers.

Hence, our assumption is wrong.

Thus, $\sqrt{2}$ is an irrational number. (1)

Concept Applied

- Co-prime numbers are pairs of numbers that do not have any common factor other than 1.

86. Suppose $2 + 5\sqrt{3}$ is a rational number.

$\therefore$ We can find two integers a, b ($b \neq 0$) such that

$$2 + 5\sqrt{3} = \frac{a}{b}, \text{ where } a \text{ and } b \text{ are co-prime integers.}$$

$$\Rightarrow 5\sqrt{3} = \frac{a}{b} - 2 \Rightarrow \sqrt{3} = \frac{1}{5} \left[\frac{a}{b} - 2 \right] \quad (1)$$

$$\Rightarrow \sqrt{3} \text{ is a rational number.}$$

$$[\because a, b \text{ are integers, so } \frac{1}{5} \left[\frac{a}{b} - 2 \right] \text{ is a rational number}]$$

But this contradicts the fact that $\sqrt{3}$ is an irrational number. (1)

Hence, our assumption is wrong.

Thus, $2 + 5\sqrt{3}$ is an irrational number. (1)

Expert Tips

- $\sqrt{p}$ is irrational if $p > 0$ and not a perfect square, i.e., $\sqrt{2}$ is irrational but $\sqrt{4} = 2$ is rational.

87. Irrational number is a number which can not be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

First, we prove that $\sqrt{5}$ is an irrational number.

Let us assume that $\sqrt{5}$ is a rational number.

Then $\sqrt{5} = \frac{a}{b}$; where a and b ($\neq 0$) are co-prime integers. (1)

Squaring on both sides, we get

$$5 = \frac{a^2}{b^2} \Rightarrow a^2 = 5b^2 \Rightarrow 5 \text{ divides } a^2$$

$$\Rightarrow 5 \text{ divides } a \quad \dots(i) \quad (1)$$

$$\Rightarrow a = 5c, \text{ where } c \text{ is an integer}$$

Again, squaring on both sides, we get

$$a^2 = 25c^2$$

$$\Rightarrow 5b^2 = 25c^2 \Rightarrow b^2 = 5c^2 \Rightarrow 5 \text{ divides } b^2$$

$$\Rightarrow 5 \text{ divides } b \quad \dots(ii)(1)$$

From (i) and (ii), we get

5 divides both a and b .

$\Rightarrow a$ and b are not co-prime integers.

This contradicts the fact that a and b are co-primes.

Hence, $\sqrt{5}$ is an irrational number.

Now, to prove $3 + 2\sqrt{5}$ is an irrational number.

Suppose $3 + 2\sqrt{5}$ is a rational number

$\therefore$ We can find two integers a, b ($b \neq 0$) such that

$$3 + 2\sqrt{5} = \frac{a}{b} \quad (\text{where } a \text{ and } b \text{ are co-prime})$$

$$\Rightarrow 2\sqrt{5} = \frac{a}{b} - 3 \Rightarrow \sqrt{5} = \frac{1}{2} \left(\frac{a}{b} - 3 \right)$$

$$\Rightarrow \sqrt{5} \text{ is a rational number} \quad (1)$$

$$\left[\because a, b \text{ are integers, } \therefore \frac{1}{2} \left(\frac{a}{b} - 3 \right) \text{ is a rational number} \right]$$

But this contradicts the fact that $\sqrt{5}$ is an irrational number.

Hence, our assumption is wrong.

Thus, $3 + 2\sqrt{5}$ is an irrational number. (1)

Alternative Method

Irrational number is a number which can not be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

First, we prove that $\sqrt{5}$ is an irrational number.

Let us assume that $\sqrt{5}$ is a rational number.

Then $\sqrt{5} = \frac{a}{b}$; where a and b ($\neq 0$) are co-prime integers.

Squaring on both sides, we get